RELEASE v0.1.0 · 2026-10-29

Cohort Social Dynamics Model

A time-aware model of friendship, groups, romance and conflict in a university cohort

- **How the model works**
- **How to put your data in**
- **The pilot test: a 10-student, 2-semester invented cohort**
- **What the model found, and how it compares with the story**

Context	AUT Mechanical Engineering, cohort entering Mehr 1403
Repository	github.com/omidanrahyabco/social-analysis-
Source	branch claude/optimistic-ride-79ocwj, commit 13f4ce7

SYNTHETIC DATA

Every person, rating and event in the pilot test is invented to test the pipeline.

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Release notes

v0.1.0: model and pilot test

This is the first release of the cohort model. It does three things:

- **The model.** It takes per-wave relationship ratings, shared contexts and dated events for a student cohort, and turns them into a description of the network over time, an explanation of why ties changed, a validated simulation and a forecast.
- **The data format.** Six CSV files that a survey can be converted into, plus a tool that turns square survey sheets into the main ratings file.
- **The pilot test.** An invented cohort of 10 students over 2 semesters, run through the full pipeline. The results are checked against the story that was written into the data.

What is in the repository

Path	Contents
`run_analysis.py`	runs the whole pipeline on one dataset folder
`cohort/io.py`	loads and validates the six files, fills skipped answers
`cohort/network.py`	per-wave measures, structural balance, groups tracked across waves, roles
`cohort/models.py`	tie-change regression with bootstrap intervals
`cohort/simulate.py`	agent-based simulation: validation and forecast
`cohort/report.py`, `cohort/report_template.html`	findings, predictions, profiles and the interactive report
`cohort/convert.py`	survey matrices to `dyads.csv`
`data/example_10p_2sem/`	the pilot's invented dataset
`examples/make_example_data.py`	the story and the noise that produced it
`outputs/example_10p_2sem/`	the pilot's report and results
`docs/`	this document's chapters as Markdown, the figures, and the script that builds this PDF

Pilot results in brief

- The pipeline ran end to end on 1,785 noisy ratings with 3 skipped answers, in about 10 seconds.
- Of 12 patterns written into the story, it found 8, found 2 partly, missed 1, and 1 could not be tested with this data.
- One step ahead, it predicts friendship ties with half the error of assuming nothing changes. Run freely from the first wave, it reproduces the overall structure but not the conflict cascade of semester 2.
- The forecast for semester 3 includes a 65% chance that P06 and P10 start dating, 45% that P03 and P09 break up, and 84% that P03 ends up with at most one friend.

Known limits

- Ten people and four waves are enough to test the machinery, not to draw conclusions.
- The tie-change regression stands in for RSiena, which should be used on the full data.
- Six rates in the simulation (side-taking, jealousy, rivalry, fights, hostility decay, balance resolution) are assumptions. The pilot suggests the side-taking and jealousy rates are too low.

Rebuilding this document

```
python run_analysis.py data/example_10p_2sem
python docs/build_release_pdf.py
```

How the model works

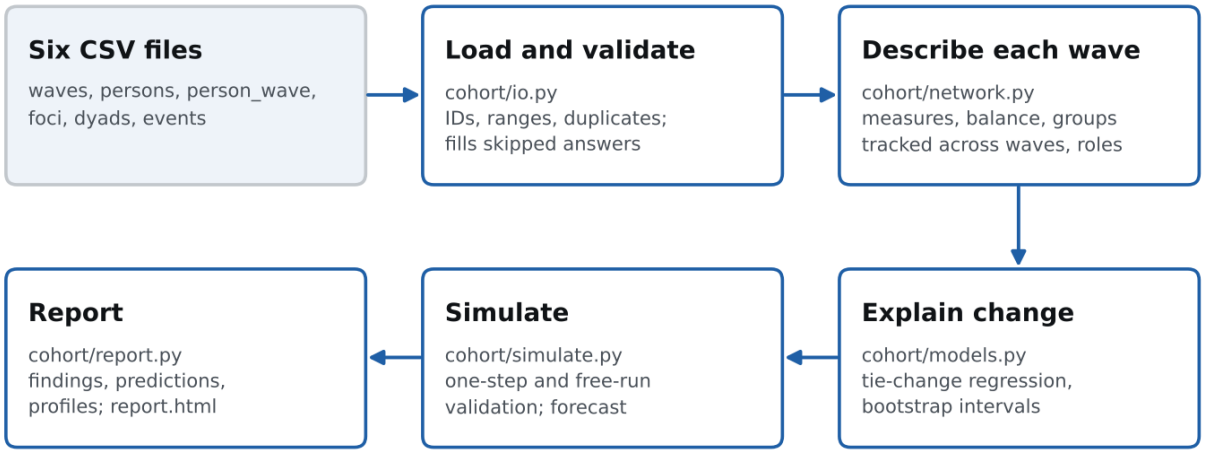
The idea

The earlier version of this project drew the cohort as one graph. A single graph cannot show that friendships form in the dorm in semester 1 and dissolve once people move out, that a breakup in semester 2 turns two friend groups against each other, or that one person thinks a friendship is close while the other does not. This model treats the cohort as a **network that changes over time, with several kinds of relationship at once, negative ties as well as positive ones, and a separate view from each side of every pair**.

It has four parts that build on each other:

Part	Question it answers	Method
A. Data model	What happened, and when?	People, contexts, five relationship layers and dated events, all per survey wave
B. Description	What did the network look like at each wave?	Snapshot measures, homophily, structural balance, groups tracked across waves, roles
C. Explanation	Why did ties form, weaken or turn hostile?	Tie-change regression (pilot); stochastic actor-oriented model (RSiena) for the full study
D. Simulation	Can we reproduce the history, and what comes next?	Agent-based model built from the fitted effects, validated on the observed waves, then run forward

A last layer reads the results against psychology and sociology theory, and a report turns everything into figures and plain-language findings.



The pipeline. Each box is one module of the code; arrows show what it hands on.

A. The data model

Waves

Time is measured in **survey waves**. For a semester structure, two waves per semester (middle and end) catch changes that one wave per semester would average away. The pilot uses four waves over two semesters. The full study has six semesters, so it would have 6 to 12 waves.

Relationship layers

At each wave, every person rates every other person on five layers. The ratings are **directed**: P04's rating of P08 and P08's rating of P04 are separate numbers. The gap between them is data, because unreturned feelings are where a lot of drama starts.

Layer	Scale	Meaning
friendship	0–5	0 do not know them, 1 acquaintance, 2 friendly, 3 friend, 4 close friend, 5 best friend
trust	0–5	would tell them something private
hostility	0–5	0 none, 1 irritation, 3 open dislike, 5 enemy
romantic	0–3	0 none, 1 interest or crush, 2 dating, 3 ex
contact	0–5	how often they actually interact, 0 never to 5 daily

Friendship and hostility are separate scales on purpose. Two people can be close and in conflict at the same time, and a single "relationship score" would hide that.

Tie definitions

The analysis turns ratings into ties with fixed rules:

- **Friendship tie**: the mean of the two friendship ratings is 3 or more.
- **Hostile tie**: the mean of the two hostility ratings is 2 or more, or either side rates 3 or more. Hostility overrides friendship.
- **Signed tie**: +1 for friendship, −1 for hostility, 0 otherwise. This is the matrix the balance analysis uses.
- **Couple**: both rate romantic = 2. **Exes**: both rate 3. **Interest**: one rates 1 (marked mutual if both do).

Contexts

A context (in sociology, a focus) is anything that puts people in the same place regularly: a dorm block, a class section, an online group, a study group, a rented flat, a student association. Membership is recorded per wave, because people move out, change section and leave chats. A context that includes everyone, such as the whole class's group chat, carries no information about who meets whom, and the loader ignores it.

Events

Dated incidents such as starting to date, a breakup, a public fight, a rejection, moving house, a group forming or going silent, or someone helping someone. Each has actors, third parties who were involved, and an intensity from 1 to 5. Event types are classed as positive or negative so the model can use them.

B. Describing each wave

Cohort measures

Measure	Definition	What it shows
Density	share of directed pairs with friendship ≥ 3	how connected the cohort is
Reciprocity	share of directed friendship ties that are returned	mutual versus one-sided friendship
Transitivity	closed triangles ÷ connected triples, on friendship ties	how clustered into groups
Friendship / hostile ties	counts of undirected signed ties	the size of each side
Tension	unbalanced triangles ÷ all fully signed triangles	how much drama the structure holds
E-I index	(ties between categories − ties within) ÷ all ties	homophily by gender, housing, origin; −1 all within, +1 all between
Context share	share of friendship ties whose two people share a context	how much the environment explains
Stability	Jaccard overlap of friendship ties with the previous wave	how much the network reshuffles
Perception gap	mean of |A→B − B→A|	how differently pairs see their relationship

Each E-I index is compared with the value random mixing would give, which depends on the group sizes. A cohort with 6 men and 4 women has a random gender E-I of +0.07, not 0.

Structural balance

Heider's balance theory says some triangles of relationships are stable and others are not. With signs on all three sides:

- **+ + +** (three friends) and **+ − −** (two friends with a shared enemy) are balanced.
- **+ + −** (one person is friends with two people who are hostile to each other) is unbalanced. The person with the two positive ties is **caught in the middle**. These triangles tend to resolve: the middle person drops one side, or the two sides reconcile.
- **− − −** is weakly unbalanced and rare.

The **tension** measure is the share of fully signed triangles of the + + − kind. The model also records who is in the middle of each one.

Groups and how they change

At each wave, groups are found with the Louvain method on friendship strength (mean rating 2 or more, hostile pairs excluded), keeping the best of 40 random starts by modularity. Groups found at different waves are then matched by member overlap (Jaccard ≥ 0.34), so a group keeps its name while most of its members stay. Each change is labelled: **formed** (and from which earlier groups), **split**, **dissolved**, **changed** (who joined, who left) or **stable**.

Roles

Each person gets one or more roles per wave from their position in the network:

Role	Rule
hub	friendship ties ≥ max(3, mean + 1 SD)
broker	among the top two in betweenness centrality, and above 0.15
caught in the middle	holds the two positive ties of a + + − triangle
in open conflict / one enemy	two or more / exactly one hostile tie
peripheral	one friendship tie or none
group member	none of the above

C. Explaining change: the tie-change model

For every directed pair (i rating j) and every pair of consecutive waves, the outcome is the change in i's friendship rating of j:

```
ΔF_ij(t→t+1) = β0 + Σ_k β_k · x_k(i, j, t) + ε
```

The explanatory variables are measured at the earlier wave, except contexts and events, which are measured over the interval:

Variable	Theory it tests
Current friendship level	state dependence: strong ties have less room to grow
j rates the tie higher than i does	reciprocity: people adjust toward the other side
Common friends	triadic closure: friends of friends become friends
Shared contexts at the next wave	propinquity and focus theory
Shared contexts × semester 2	do contexts matter less once everyone knows everyone?
Lost a shared context	moving out, changing section
Same gender, same housing type	homophily
GPA gap	academic homophily
Attractiveness of j	attraction and status
Hostility between them	conflict erodes friendship
Unbalanced triangles through the tie	balance pressure
Negative / positive events between them	the shock of a breakup, fight, new relationship or help

The model is fitted by ordinary least squares on standardized variables, so each effect reads as "rating points per standard deviation". Uncertainty comes from 2,000 bootstrap resamples over pairs: both directions and all waves of a pair are drawn together, because they are not independent. An effect counts as **clear** when its 90% interval does not cross zero.

This regression is a deliberately simple stand-in. On the full 30-person dataset the standard method is the **stochastic actor-oriented model (SAOM, R package RSiena)**. It models tie changes as choices people make, handles the dependence between ties properly, and can model network and behaviour (for example GPA) changing together. The variables above map directly onto RSiena effects (reciprocity, transitive triplets, same-covariate, covariate-alter, dyadic covariates), so the pilot's design carries over.

D. Simulation

The simulation is an agent-based model. Each person is an agent carrying their attributes and current ratings. One simulation step is one survey wave, and it runs in this order:

1. **Breakups.** Each couple breaks up with probability p_break, which doubles while either partner is interested in someone else. A breakup sets both to "ex" and adds hostility, more on the side that was hurt. Close friends of one partner (rating 4 or more, and clearly closer to that side) take sides with probability 0.40.
2. **New couples.** Mutual interest between two single people becomes a relationship with a probability estimated from the data. Extraverts are more likely to act on it and introverts less. An ex of either person may react with hostility toward the new partner (probability 0.60).

3. **Interest.** One-way interest is returned with a probability that rises with the admirer's attractiveness and the existing friendship. Otherwise it fades. New interest appears between people who meet, at a rate estimated from the data, and less often toward someone who is taken.
4. **Escalation.** Pairs who already dislike each other and still meet may fight, which adds hostility and counts as a negative event.
5. **Friendship.** Every rating moves by the fitted tie-change model plus noise. The noise is partly shared by both directions of a pair, so a pair's two ratings tend to move together.
6. **Fading.** Hostility decays by 15% per wave unless something feeds it.
7. **Balance.** In each unbalanced triangle, the person in the middle drifts away from one side with probability 0.30.

Where the numbers come from

Rate	Source
All friendship effects and the noise level	fitted tie-change model
Breakup hazard	event log: breakups ÷ months in a relationship, with a Gamma(1, 12) prior (about one breakup per couple-year)
New interest	romantic layer: new interest ÷ opposite-gender pairs at risk, with a Beta(1, 30) prior
Mutual interest → couple	romantic layer, with a Beta(2, 4) prior
Side-taking 0.40, jealousy 0.60, rivalry 0.35, fights 0.15, hostility decay 0.85, balance resolution 0.30	assumptions for the pilot; to be estimated from the full data

The priors keep a tiny dataset from producing extreme rates. In the pilot, two of two mutual interests became couples; without a prior the model would assume this always happens.

Validation

Two tests run on the observed waves before any forecast is trusted:

- **One-step-ahead.** Start from each real wave, simulate the next one 300 times, and score the predicted probability of every friendship tie against what was observed, using the Brier score (mean squared error of a probability; lower is better). The baseline assumes every tie stays exactly as it was.
- **Free run.** Start from the first wave and let the model generate everything after it, including the romance and conflict, 500 times. Then check whether the observed cohort measures fall inside the middle 80% of simulated histories.

Forecast

The forecast starts from the last observed wave and runs a given number of waves ahead (two by default), 1,000 times. Each run gets its own plausible future contexts: homes, dorms and flats stay as they are, class sections are reshuffled at random, the study group continues with probability 0.75 and the association with probability 0.70. The outputs are probabilities: each pair being friends or hostile at the end, each pair dating at some point, current couples surviving, unbalanced triangles resolving one way or the other, each person's expected number of friends and isolation risk, the most likely group structure, and the range of each cohort measure.

Psychological and sociological reading

Pattern	Theory
First friendships form inside dorms, sections and chats	Focus theory (Feld 1981); propinquity (Festinger, Schachter & Back 1950)
People befriend people like them; this weakens with contact	Homophily (McPherson, Smith-Lovin & Cook 2001); contact hypothesis (Allport)
A tie moves toward how the other side rates it	Reciprocity norm (Gouldner 1960); social exchange
Friends take sides after a breakup	Structural balance (Heider 1946; Cartwright & Harary 1956)
People between groups gain visibility and risk	Structural holes (Burt 1992); weak ties (Granovetter 1973)
Losing your context means losing friends	Dunbar layers; the cost of maintaining ties
Unequal investment in a relationship	Perception asymmetry; attachment
Groups form, clash, then settle	Group development (Tuckman 1965)

From the pilot to the full study

1. Collect the real data in the format in the data guide: 30 people, 6 semesters, ideally two waves per semester.
2. Run the pipeline unchanged first. Every descriptive output works at any size.
3. Fit RSiena on the friendship and hostility layers with the same effects, including time-dummy effects to test whether contexts matter less after year 1.
4. Fit a relational event model on the event log to model cascades (a breakup followed by side-taking followed by fights) directly.
5. Re-estimate the simulation's rule rates from the real data, and check the validation tests again before reading the forecast.

How to put your data in

Folder and command

One dataset is one folder with six CSV files (UTF-8, comma-separated, header row first):

```
data/aut_mech_1403/
  waves.csv
  persons.csv
  person_wave.csv
  foci.csv
  dyads.csv
  events.csv
```

Then run:

```
pip install -r requirements.txt
python run_analysis.py data/aut_mech_1403
```

The results go to `outputs/aut_mech_1403/`: `report.html` (open it in a browser) and `results.json` (every number in the report). On the 10-person pilot a full run takes about 10 seconds. Add `--runs 2000` for a steadier forecast or `--steps 4` to look further ahead.

IDs and privacy

- Give every person a pseudonym such as `P01` to `P30`, and use exactly the same IDs in every file.
- Keep the file that links names to IDs on one offline machine, never in this repository.
- A pseudonym is not enough on its own in a cohort everyone knows. "The girl from Rasht in dorm B with the top GPA" identifies a person. Record origin in broad bands (Tehran / other city, or a region) when that is enough for the question.
- Get informed consent before collecting the romantic and hostility layers, which are the most sensitive. Decide in advance what you will do with answers people give about others who did not consent.

waves.csv

One row per survey wave, in time order.

Column	Example	Notes
wave	`W1`	short ID used in every other file
label	`S1 mid (Aban 1403)`	shown in the report
date	`2024-11-15`	ISO date the wave refers to; events are assigned to the interval between waves
semester	`S1`	waves in the same semester share this; the model compares later semesters with the first

persons.csv

One row per person. Attributes that do not change.

Column	Example	Notes
id	`P07`	pseudonym
gender	`F` / `M`	

Column	Example	Notes
origin	`Tehran`, `Shiraz`	anything that is not `Tehran` counts as "other city" in the homophily measure
entry_age	`18`	
extraversion, agreeableness, conscientiousness, neuroticism, openness	`4.2`	Big Five, 1–5, from a short questionnaire such as the BFI-10
attractiveness_peer	`3.6`	1–5, the **mean of several peers' ratings**, never one person's opinion

person_wave.csv

One row per person per wave.

Column	Example	Notes
id	`P02`	
wave	`W3`	
housing	`family`, `dorm`, `rented`	keep the same few words throughout
gpa	`18.2`	out of 20; repeat the semester GPA on both waves of a semester
status	`active`	`active`, `left` or `graduated`

foci.csv

One row per person, context and wave. A context is any place or group that puts people together regularly.

Column	Example	Notes
id	`P05`	
focus	`dorm-A`, `dynamics-sec1`, `study-group`	a name for the context
type	`dorm`, `class`, `online`, `study`, `rented`, `extracurricular`	the forecast keeps dorms and flats, reshuffles classes, and keeps study and extracurricular groups with some probability
wave	`W3`	list the person again at every wave they are in it

A context that includes the whole cohort, such as the class-wide group chat, is ignored automatically.

dyads.csv

The core file: one row per person, per person they rate, per wave, per layer.

Column	Example	Notes
rater	`P04`	who answered
target	`P08`	who they answered about
wave	`W3`	
layer	`friendship`	one of `friendship`, `trust`, `hostility`, `romantic`, `contact`
value	`4`	whole number in the layer's range

Anchors to print on the questionnaire:

Layer	0	1	2	3	4	5
friendship	do not know them	acquaintance	friendly	friend	close friend	best friend
trust	none			would share a private matter		complete
hostility	none	irritation		open dislike		enemy
romantic	none	interest or crush	dating	ex		
contact	never	rarely	monthly	weekly	several times a week	daily

For 30 people, 6 waves and 5 layers this file has 30 × 29 × 6 × 5 = 26,100 rows. Nobody should type that by hand. Collect each layer and wave as a square sheet (first column: the person answering; header row: the people being rated), save each sheet as `<layer>_<wave>.csv`, for example `friendship_W1.csv`, and convert the whole folder:

```
python -m cohort.convert path/to/matrices data/aut_mech_1403/dyads.csv
```

Leave a cell empty when someone skipped it. The loader fills a skipped rating with the other person's rating of the same tie and reports how many it filled.

events.csv

One row per dated incident.

Column	Example	Notes
date	`2025-04-06`	the month is enough; use the 15th if unsure
type	`breakup`	see below
actors	`P03;P08`	the people the event is between, separated by `;`
third_parties	`P09;P04;P10`	others who were pulled in; may be empty
intensity	`5`	1 minor to 5 life-changing
note	`right after Nowruz`	optional, free text, no names

Recognized types:

- **Negative:** breakup, fight, public_fight, cold_shoulder, crush_rejected, gossip, group_dissolved, betrayal
- **Positive:** started_dating, group_formed, support, moved_housing, reconciliation

Other types are kept in the report but not used by the model.

Collecting three years after the fact

The cohort is three years in, so most data will be recalled, not observed. Some ways to make recall as reliable as possible:

- **Life-history calendar.** Give each person a timeline with the semesters, exam periods, Nowruz and summers marked, and ask them to place housing moves, relationships and fights on it first. Rate the dyads afterwards, one wave at a time, with the calendar in front of them.
- **Ask both sides.** Every tie is rated by both people. Where they disagree strongly about an event (one says dating, the other says nothing), check the event log and mark the tie as uncertain rather than picking one side.

- **Start with the present.** Recent waves are remembered best. Rate the latest wave first and work backwards.
- **Rate one layer per page.** Mixing friendship and hostility on one page invites people to give one overall impression.

Checklist before running

- ■ Every ID in every file appears in `persons.csv`.
- ■ Every wave in every file appears in `waves.csv`.
- ■ Values are in range: 0–5, or 0–3 for romantic.
- ■ Nobody rates themselves (self-ratings are dropped with a warning).
- ■ Each person appears in `person_wave.csv` at every wave.
- ■ Event dates fall between the first and last wave dates, or they are ignored by the model.

The loader stops with a clear message on errors (unknown IDs, unknown layers, out-of-range values) and warns about things it can fix (duplicates, self-ratings, skipped answers). Warnings are listed at the top of the report.

The pilot test

Why a fake cohort

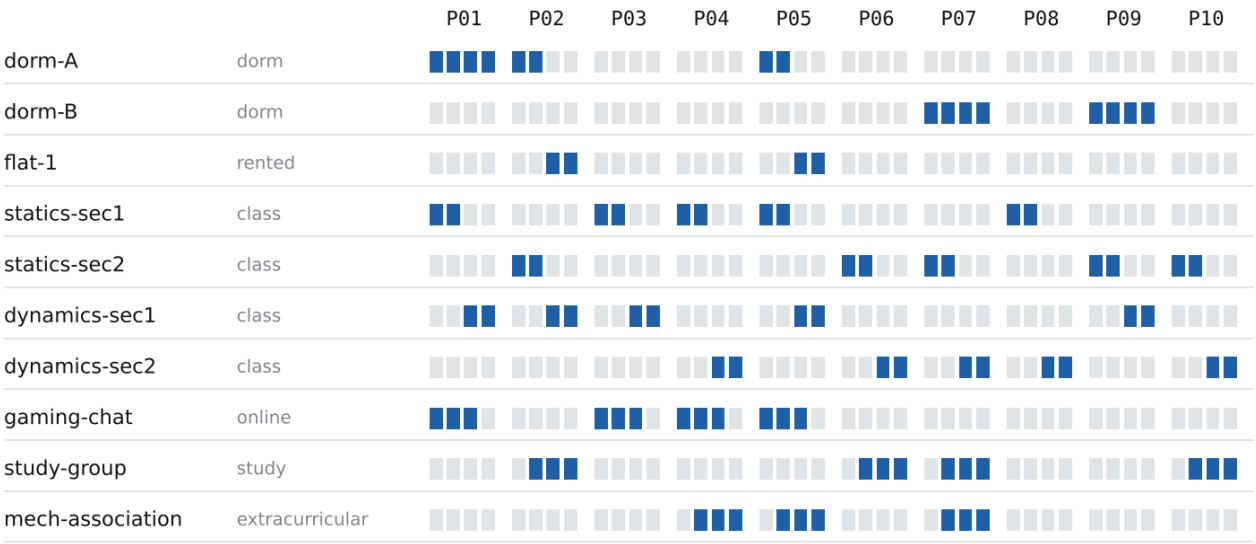
Before real data from 30 people is collected, the pipeline needs a test where the right answer is known. So we wrote a story about 10 invented students over their first two semesters, turned it into the survey answers those students would plausibly have given, fed only those answers to the pipeline, and checked what it found against what we had planted.

All people, ratings and events in this test are invented. The data is in `data/example_10p_2sem/` and the generator is `examples/make_example_data.py`. The pipeline never reads the generator, only the six CSV files.

The invented cohort

Ten students enter AUT mechanical engineering in Mehr 1403 (September 2024): six men and four women. Five are from Tehran and live with family; the others came from other cities and live in the dorms, men in block A and women in block B.

ID	Gender	From	Housing	GPA S1 → S2	Personality	Role in the story
P01	M	Isfahan	dorm	15.1 → 13.8	very extraverted	the early connector who is left behind
P02	M	Tabriz	dorm → rented flat	17.9 → 18.2	introverted, conscientious	studious; moves out with P05
P03	M	Tehran	family	14.2 → 13.1	extraverted, low agreeableness, most attractive man	at the centre of both romances
P04	M	Tehran	family	15.6 → 15.9	agreeable, calm	P03's best friend who ends up on the other side
P05	M	Mashhad	dorm → rented flat	15.0 → 15.4	agreeable, stable	the bridge; quietly interested in P09
P06	M	Tehran	family	18.6 → 18.9	very introverted, top GPA	study group; slowly drawn to P10
P07	F	Shiraz	dorm	17.4 → 17.8	extraverted, agreeable	study group; rooms with P09; caught in the middle
P08	F	Tehran	family	15.4 → 14.6	most attractive, highest neuroticism	dates P03; the breakup
P09	F	Rasht	dorm	14.8 → 14.5	attractive	P03's next girlfriend
P10	F	Tehran	family	18.1 → 18.4	introverted, conscientious	P08's best friend; study group



Contexts in the story: dorms, class sections, the gaming chat, the study group and the association, per wave.

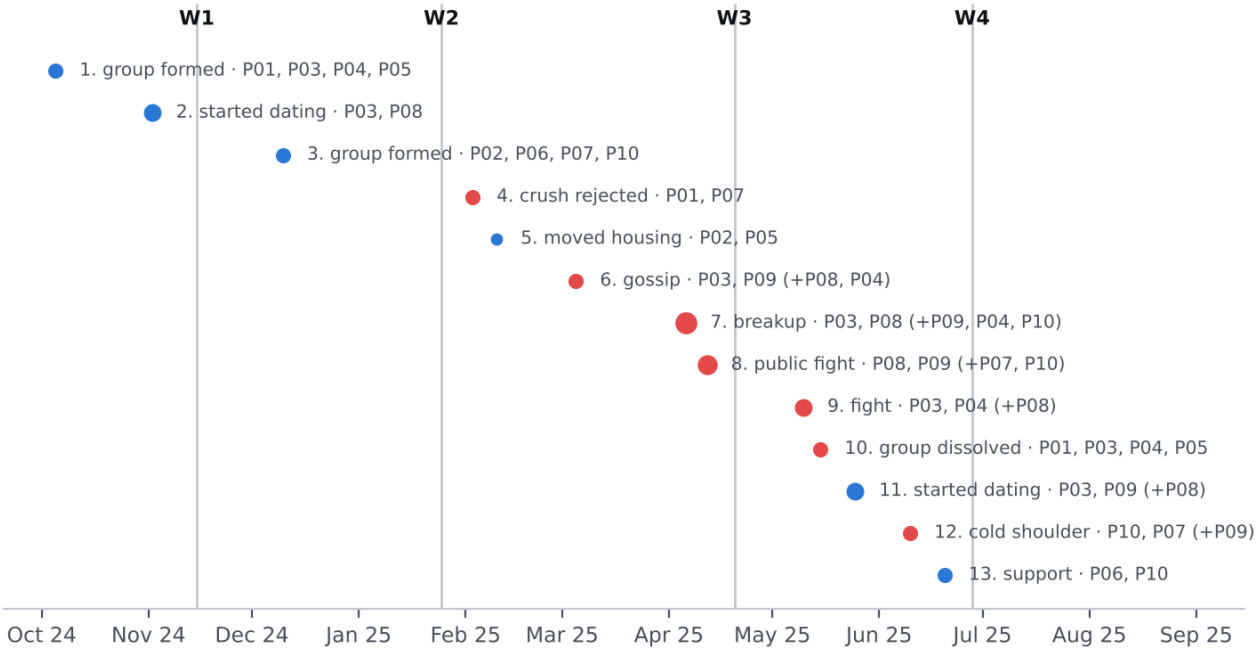
The story

Semester 1, first weeks. Friendships form where people meet. The dorm-A men (P01, P02, P05) become a unit. The Tehran men in statics section 1 (P03, P04) start a gaming chat with P01 and P05, which makes P01 a hub between the dorm and Tehran circles. P07 and P09 share a room in dorm B. P08 and P10 are friends from home. P03 and P08 are interested in each other.

Semester 1, later. P03 and P08 start dating in November. Before the midterms the strongest students (P02, P06, P07, P10) form a statics study group, the first friendships that cross the gender line. P01 develops a crush on P07.

Between semesters. P07 turns P01 down in February. The same week P02 and P05 leave the dorm and rent a flat together. P01 stays behind in dorm A, alone.

Semester 2. The class sections are reshuffled. P03 and P09 land in the same dynamics section and are seen together often; the gossip reaches P08 in March. Right after the Nowruz break P03 and P08 break up (intensity 5). Within a week P08 and P09 have a public fight. P10 sides with her best friend P08. P04, now in P08's section, defends her, and fights with P03 in May. The gaming chat goes silent. P03 and P09 start dating at the end of May. P10 gives P07 the cold shoulder for still rooming with P09. Meanwhile, P06 helps P10 prepare for exams, and by the end of the semester they are interested in each other. P05 has a crush on P09 and has told no one.



The event log as the pipeline received it. Numbers match the event table in the report.

From story to survey answers

Real survey answers are noisy, so the generator does not simply copy the story. For every rating it adds:

- **personal rating style:** extraverts rate everyone a little higher (+0.25 per point of extraversion above 3);
- **recall noise:** about one rating in five is off by one point in either direction;
- **perception gaps:** some people see a friendship as closer than the other side does (P01 about his old dorm-mates, P04 about P08, P09 about P07);
- **stray irritations:** a 3% chance of a hostility rating of 1 that the story never planned;
- **skipped answers:** P06 skipped two ratings at W1 and P03 skipped one at W3.

Trust and contact are derived from friendship, hostility, agreeableness and shared contexts, with their own noise. The result is 1,785 ratings, 13 events, 133 context memberships and 40 person-wave records.

The run

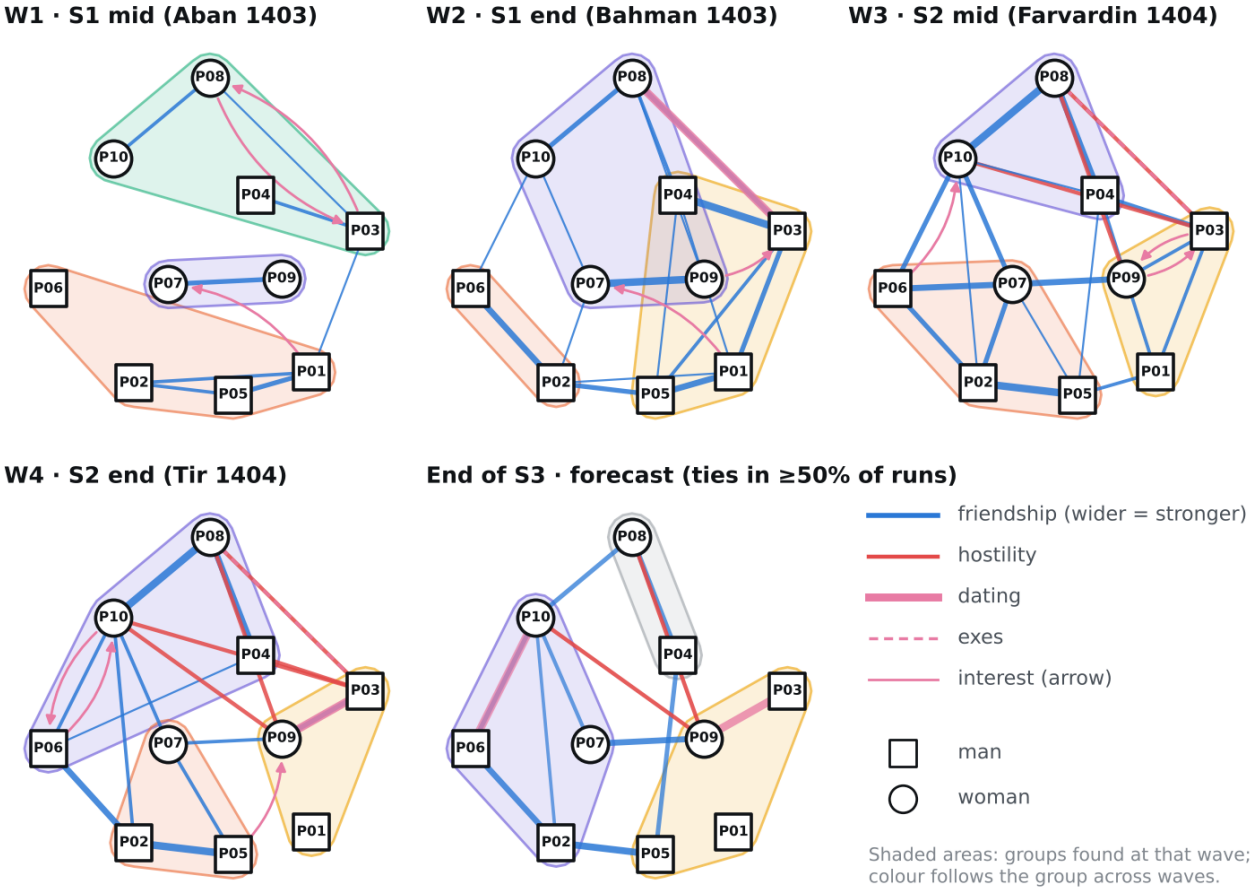
```
python run_analysis.py data/example_10p_2sem
```

```
[warning] dyads: 3 skipped answers imputed from the other person's rating of the same tie
loaded 10 people, 4 waves, 1785 ratings, 13 events
tie-change model: R2=0.49 on 270 pair-transitions
    65%  P06 and P10 start dating
    45%  P03 and P09 break up
    29%  P07 sides with P09 and drifts from P10
    84%  P03 ends S3 with at most one friend
    53%  P01 ends S3 with at most one friend
    43%  New hostility between P03 and P09
    49%  Existing hostilities last through S3
```

The validator found the three skipped answers and filled them. A full run (2,000 bootstrap samples, 300 runs per one-step test, 500 free runs, 1,000 forecast runs) takes about ten seconds.

What the pipeline saw

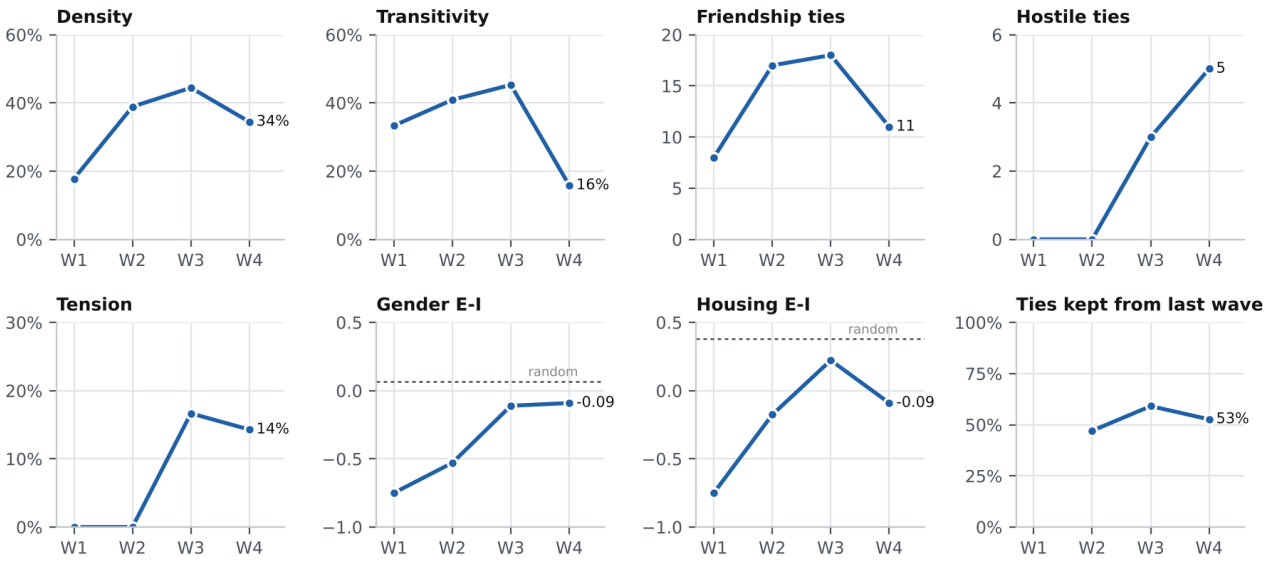
The network, wave by wave



The cohort at each wave and the forecast for the end of semester 3. Blue: friendship, wider for stronger. Red: hostility. Pink: dating (thick), exes (dashed), interest (arrow). Shaded areas: groups. Squares are men, circles women.

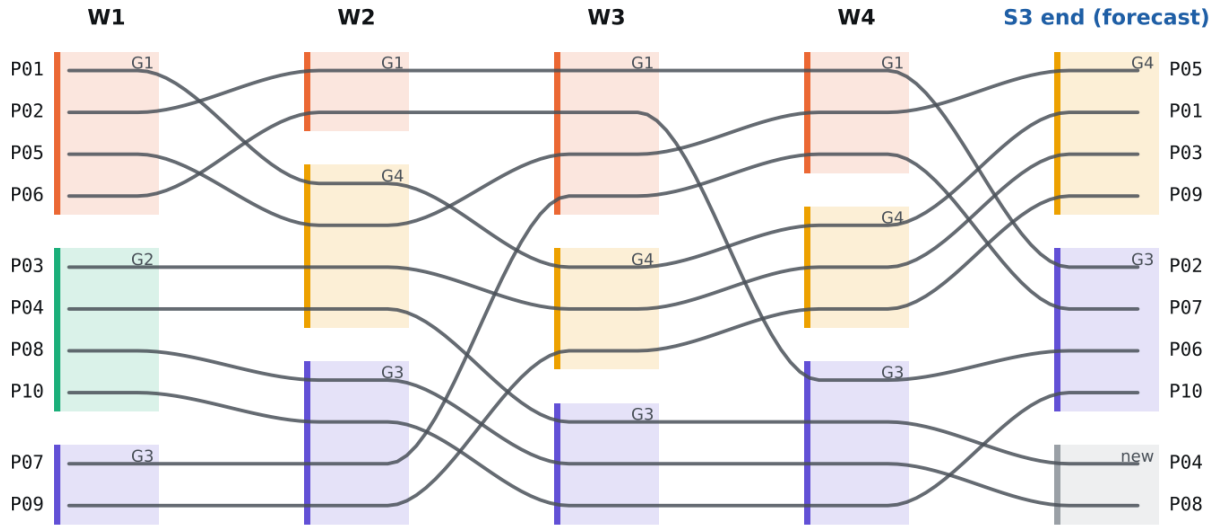
	W1	W2	W3	W4
Friendship ties	8	17	18	11
Hostile ties	0	0	3	5
Density	18%	39%	44%	34%
Transitivity	33%	41%	45%	16%
Tension	0%	0%	17%	14%
Gender E-I (random: +0.07)	−0.75	−0.53	−0.11	−0.09
Ties inside a shared context	88%	88%	100%	100%
Ties kept from previous wave		47%	59%	53%
Couples	0	1	0	1

The cohort grows denser through semester 1, peaks at W3, then contracts sharply: eight friendship ties are lost between W3 and W4 and only one is gained. Triangles of friends break up (transitivity falls from 45% to 16%) as hostility spreads.



Cohort measures over the four waves. Dotted lines: what random mixing would give.

Groups



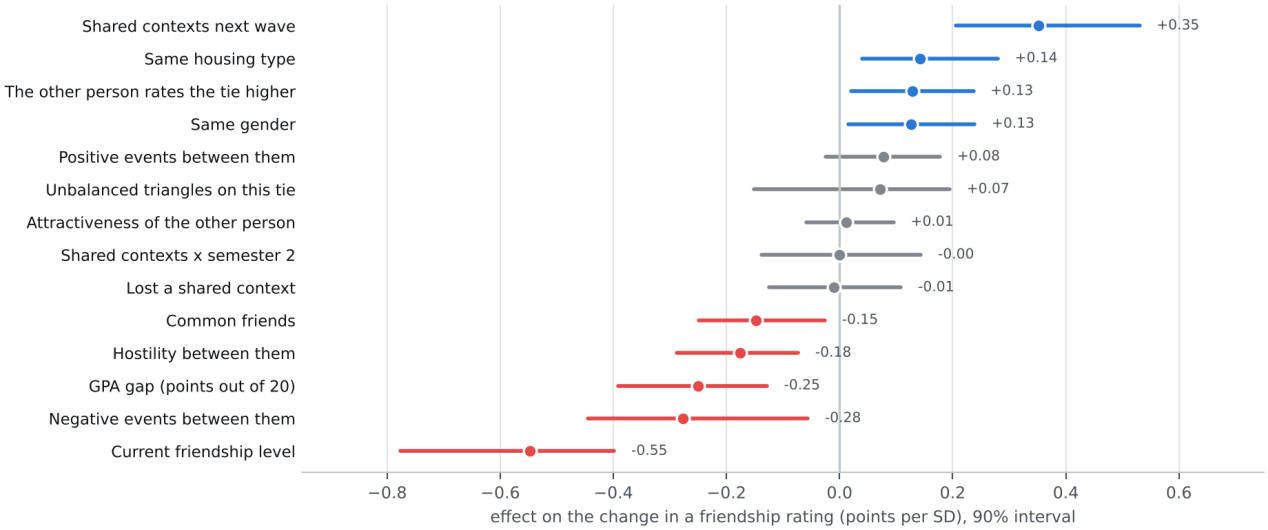
Each line is one person; shaded bands are the groups found at each wave, with the forecast on the right.

- **W1:** three groups that follow housing and home city: the dorm men with P06 (G1), the Tehran circle of P03, P04, P08 and P10 (G2), and the dorm-B roommates (G3).
- **W2:** the first-semester groups split. The gaming chat pulls P01, P03, P04 and P05 into a new G4. The women merge into one group (G3). P02 and P06 are left as the study-group core (G1).
- **W3:** after the breakup the groups re-sort along the conflict. P03 and P09 are grouped with P01; P04 joins P08 and P10.
- **W4:** P04 is now firmly with P08, P10 and P06. P01, P03 and P09 form the "new couple and the left-behind" group. P02, P05 and P07 form the flat-and-association group.

Roles

P01 and P03 are both hub and broker at W1. By W4, P01 is peripheral with no friendship ties, and P03 is in open conflict with three people. P04 is caught in the middle at W3 (between P03 and P08). P07 is caught in the middle at W4 (between P09 and P10). P10 goes from peripheral at W1 to hub and broker at W4.

Why ties changed



Effects on the change in a friendship rating between waves, in rating points per standard deviation, with 90% bootstrap intervals.

Clear effects:

- **Shared contexts** (+0.35): the strongest positive force.
- **Similar GPA** (gap −0.25): the study group is visible in the numbers.
- **The other person rates the tie higher** (+0.13): ratings move toward each other.
- **Same housing type** (+0.14) and **same gender** (+0.13).
- **Negative events** (−0.28) and **hostility** (−0.18).
- **Current level** (−0.55): high ratings tend to fall back, low ones to rise.
- **Common friends** (−0.15): negative, the opposite of what triadic closure predicts. With 10 people this most likely reflects the collapse of whole friend triangles at W4, not a real effect.

No clear effect: attractiveness, losing a context, positive events, unbalanced triangles, and shared contexts × semester 2.

Planted versus found

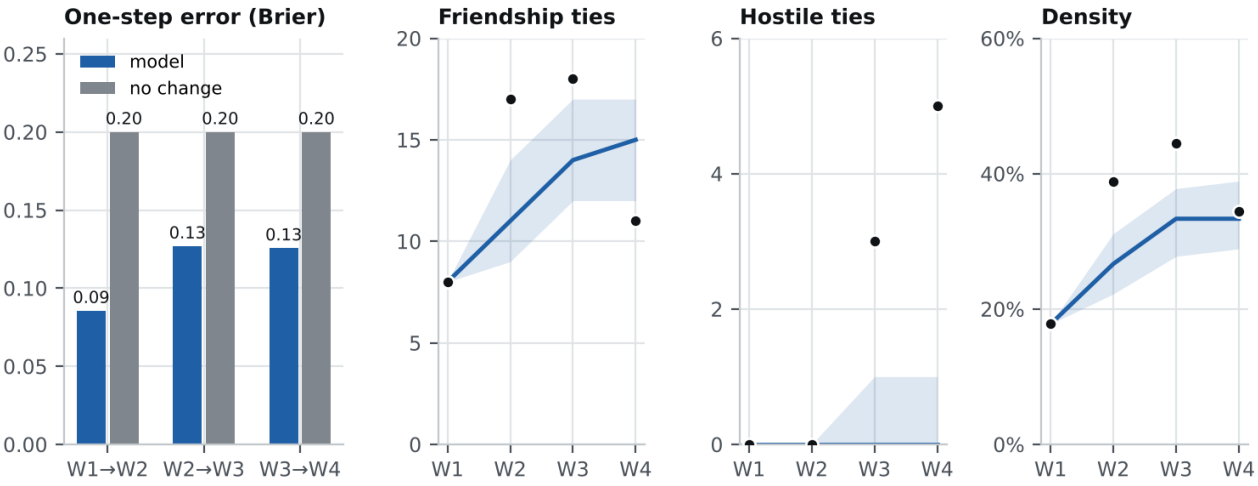
This is the real test: for each thing written into the story, did the pipeline find it from the survey answers alone?

Planted in the story	What the pipeline found	Verdict
First friendships come from dorms, sections and chats	88% of W1 ties share a context; shared context is the strongest positive effect (+0.35)	Found
Gender-separated start, mixing through the study group and romance	Gender E-I from −0.75 to −0.09, near random (+0.07)	Found
P03–P08 relationship and breakup	Couple at W2, exes from W3, hostility P08→P03; negative events −0.28	Found
Side-taking after the breakup	Hostile ties 0 → 3 → 5: P08–P09, P09–P10, P03–P04, P03–P10	Found
P04 between his best friend and P08	P04 "caught in the middle" at W3, hostile with P03 at W4	Found
P07 between roommate P09 and study partner P10	Unbalanced triangle at W4 with P07 in the middle; forecast 57% still stuck	Found

Planted in the story	What the pipeline found	Verdict
P02 and P05 move out and P01 is left behind	P01 from hub (3 ties) to 0 ties; 53% isolation risk forecast	Found, but the "lost a context" effect itself is near zero: the model sees the loss through other variables
P01 invests more than he gets back	Largest perception gaps: P01 rates P02 and P03 at 3, gets 1 back	Found
Study group P02, P06, P07, P10	Visible in the GPA effect and in P02–P06 staying together; never detected as one group, because the women's own ties were stronger	Partly found
P06–P10 mutual interest at the end	65% chance they start dating in S3	Found as a forecast
P05's secret crush on P09, who is taken	Recorded as one-way interest, but almost no consequence in the forecast (P05–P09 couple 2%, no rivalry with P03)	Missed
Contexts matter less in semester 2	No effect (0.00); every W3 and W4 friendship still shared a context, so the story did not really contain this	Not testable with this data

Of the twelve planted patterns, the pipeline found eight, found two partly, missed one, and one could not be tested with this data.

Does the model hold up?



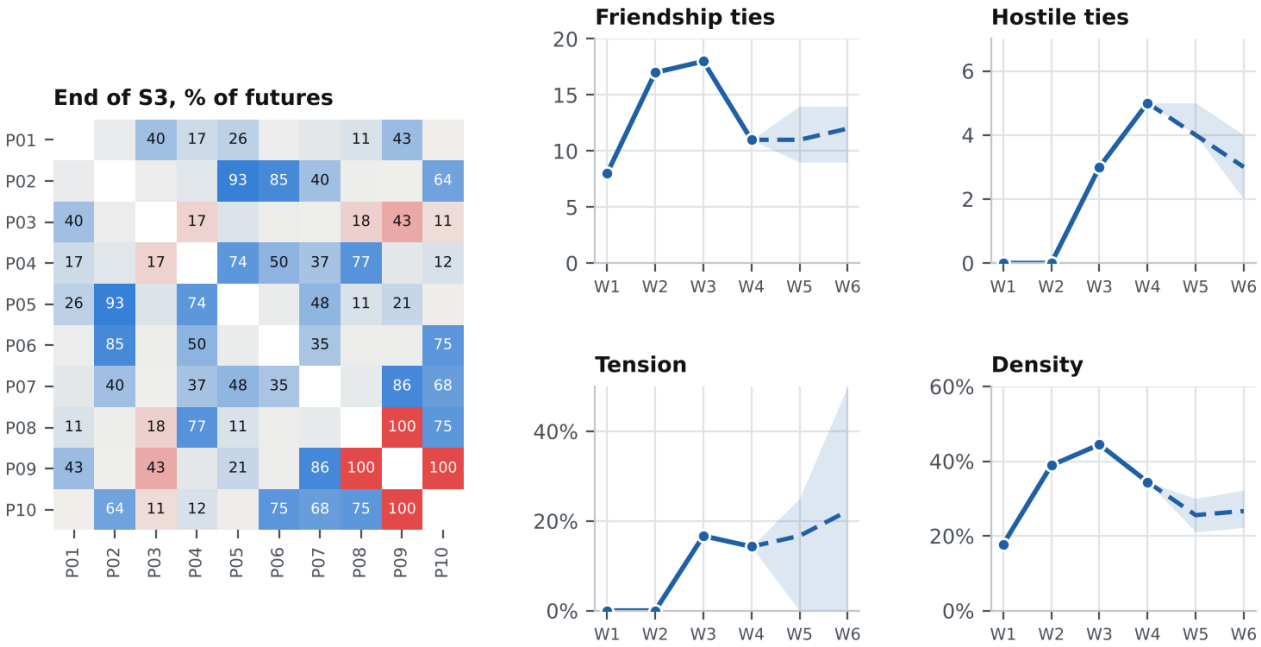
Left: one-step error on friendship ties (lower is better). Right: free run from W1 against what was observed.

One step ahead, the model halves the error of assuming no change: Brier 0.09, 0.13 and 0.13 against 0.20 for the no-change baseline at each step. But of the 27 friendship ties that appeared or disappeared, it put more than 50% on the right outcome for only 8. It knows which ties are fragile better than it knows which ones will actually flip.

Run freely from W1, it reproduces density, transitivity, gender mixing, tension and the number of couples within its 80% range at W4. It misses three things: it predicts 15 friendship ties where there were 11, a mean friendship of 2.38 where there was 1.99, and almost no hostile ties where there were 5. In the free run the breakup happens in about four runs out of ten, and when it does, the side-taking stays mostly below the hostility threshold. The cascade that drove the real semester 2 is a specific chain of events, and the model, started a year earlier, does not reproduce it. This is the main weakness to fix with real data: the side-taking and jealousy rates are

assumptions, and they appear to be too weak.

Forecast for semester 3



Left: probability that each pair is friends (blue) or hostile (red) at the end of semester 3. Right: cohort measures, observed and forecast.

Chance	Prediction
65%	P06 and P10 start dating
45%	P03 and P09 break up; 43% that the two become hostile
84%	P03 ends semester 3 with at most one friend
53%	P01 ends semester 3 with at most one friend
57% / 29% / 11%	P07 is still caught in the middle / sides with P09 / sides with P10
100%	P08–P09 and P09–P10 stay hostile

The most likely group structure at the end of semester 3 has three parts: P01, P03, P05 and P09; P02, P06, P07 and P10; and P04 with P08.

What we learned for the real study

1. **The pipeline works end to end** on data shaped like the real survey, including skipped answers.
2. **Structure is easier than drama.** Contexts, homophily and group change were found reliably. Conflict depends on specific events, so the event log matters as much as the ratings. Collect it carefully.
3. **Two waves per semester are worth it.** Most of the story happened between waves of the same semester.
4. **The rule rates need data.** Side-taking, jealousy and hostility decay are guesses in this pilot, and the free run shows the guesses are too low. With 30 people and 6 semesters they can be estimated.
5. **Ten people is too few for firm effects.** The negative common-friends effect shows how a small sample can produce a misleading sign. Read the pilot's numbers as a test of the

machinery, not as findings.